

1. Administrative & Study Identification

Full Study Title: A Phase IIb Randomised, Double-blind, Placebo-controlled Study to Evaluate the Efficacy, Safety and Tolerability of Co-administration of AZD9550 and AZD6234 in Participants Living With Obesity or Overweight With Co-morbidity (ASCEND) **Short Title/Acronym:** ASCEND **Protocol Number:** D8460C00004 **NCT Number:** NCT06862791 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD9550 and AZD6234 **Phase of Development:** Phase IIb **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine whether treatment with AZD9550 when given in combination with AZD6234 as once weekly subcutaneous (SC) injections is superior to placebo or either agent administered as monotherapy for weight loss. **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics of AZD9550 and AZD6234 as combination therapy or monotherapy.

2.2 Background & Rationale To address the critical need for effective pharmacological interventions in adults who are living with obesity or overweight and possess at least one weight-related co-morbidity (such as hypertension, dyslipidemia, or obstructive sleep apnoea). This study evaluates if a combined mechanistic approach using AZD9550 and AZD6234 delivers superior weight-loss efficacy compared to placebo or either therapy alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Active Comparator (Monotherapy Arms) **Blinding:** Double-blind **Randomization:** Randomised, parallel-group, reduced factorial

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 360 participants **Number of Sites:** Around 60 sites across approximately 7 countries **Estimated Study Start:** Q1 2025 **Estimated Primary Completion:** ~47 weeks per participant

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant must be 18 to 75 years of age inclusive. 2. BMI: ≥ 30 kg/m², or ≥ 27 kg/m² with at

least one weight related comorbidity. 3. A stable, self-reported body weight for 3 months prior to screening.

4.2 Exclusion Criteria 1. History of any clinically important disease or disorder, including GI, renal, or hepatic disease, or previous/planned bariatric surgery. 2. History of T1DM or T2DM, HbA1c $\geq 6.5\%$, or obesity induced by endocrine disorders (e.g., Cushing's syndrome). 3. Treatment with a GLP1 containing preparation within the 3 months or 5 half-lives of the drug prior to screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered as once weekly injections. **Route of Administration:** Subcutaneous (SC). **Duration of Treatment:** 47 weeks total study duration (screening, treatment period, and follow-up).

5.2 Concomitant & Prohibited Therapy Permitted: Standard treatments for existing co-morbidities (hypertension, dyslipidemia) that do not interfere with weight-loss endpoints. **Prohibited:** GLP-1 containing preparations or other weight-loss devices/pharmacotherapy during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to Day 0 Informed Consent, Medical History, BMI evaluation, Pregnancy Tests Baseline
Day 0	Randomization, First Dose Administration	Interim	
Routine study visits	Efficacy Assessment (Weight), Safety Labs, PK Sampling	End of Study	Week 47 Final Efficacy Assessment, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percent change in body weight / absolute weight loss from baseline. **Measurement Method:** Standardized clinical scale assessments during site visits.

7.2 Secondary & Exploratory Endpoints 1. Safety and tolerability profiles (Adverse Event frequencies). 2. Pharmacokinetics (PK) of AZD9550 and AZD6234 co-administration.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard reporting mechanisms during routine clinical visits and follow-up. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data**

Monitoring Committee (DMC): Internal safety monitoring managed by AstraZeneca.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary weight-loss outcomes. Safety Set for AE/SAE evaluations.
Sample Size Justification: $N=360$ appropriately powered ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect statistically significant superiority in weight loss against placebo and monotherapies.
Handling of Missing Data: Last Observation Carried Forward (LOCF) or Mixed-Model Repeated Measures (MMRM) for continuous weight endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Regular onsite and remote monitoring per sponsor protocol. **Audit Trail:** eCRF locking and data clarification forms aligned with standard regulatory requirements.

11. Study Identifiers & Governance

Primary ID: D8460C00004 **Registry Number:** NCT06862791 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ASCEND **Official Regulatory Title:** A Phase IIb Randomised, Double-blind, Placebo-controlled Study to Evaluate the Efficacy, Safety and Tolerability of Co-administration of AZD9550 and AZD6234 in Participants Living With Obesity or Overweight With Co-morbidity

12. Clinical Framework (Obesity Specific)

Primary Condition: Obesity or Overweight with Co-morbidity
Diagnosis Threshold: BMI ≥ 30 kg/m², or ≥ 27 kg/m² with comorbidity
Medical Methodology: Subcutaneous pharmacological treatment (AZD9550 & AZD6234) **Optimization Target:** Absolute and percentage body weight reduction

13. Study Procedures & Methodology

Management Pathway: Once-weekly subcutaneous injection clinical pathway
Education Protocol (HCP): Clinical trial protocol & SC administration training
Education Protocol (Patient): Self-administration guidelines & standard diet/lifestyle counseling
Quality Audit Mechanism: Clinical monitoring & patient compliance tracking

14. Observation & Follow-Up Schedule

Total Duration: 47 weeks **Onsite Visit Cadence:** Regular intervals during the 47-week timeframe **Remote Monitoring Frequency:** Intermittent phone or remote check-ins as defined by protocol
Checklist Review Interval: Routine symptom and AE check-in per clinical phase schedule

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					